STUDENT ADMISSION ENQUIRY MANAGEMENT SYSTEM USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026



RAJALAKSHMI ENGINEERING COLLEGE
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BONAFIDE CERTIFICATE

Certified that this project report titled “Student Admission Enquiry Management System Using Zoho Creator” is the bonafide work of Harrish S (2116241501501), who carried out the individual project work on Zoho Creator for CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, Harrish S (2116241501501), declare that this report titled "Student Admission Enquiry Management System Using Zoho Creator" documents the application developed in my Zoho Creator account for CB23F35 – Low-Code Application Development using Zoho Creator. It describes the design and implementation of the application and the testing carried out on it.

Artificial intelligence tools were used to assist with code suggestions and with the preparation of this report. This assistance is acknowledged. I am responsible for reviewing the work, understanding its logic and demonstrating the application.

Place: Chennai SIGNATURE OF THE STUDENT

Date: _____ Harrish S

2116241501501

ACKNOWLEDGEMENT

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I thank the Head of the Department of Artificial Intelligence and Machine Learning for the academic support provided during the course.

I thank the course faculty for guidance on the course and on the project report requirements.

I acknowledge Zoho Corporation for Zoho Creator and its documentation, which made it possible to build the application. I also thank my family and friends for their support.

ABSTRACT

The Student Admission Enquiry Management System is a low-code application developed on the Zoho Creator platform to simplify the collection and management of student admission enquiries. Colleges that record enquiries in paper registers and spreadsheets find it difficult to maintain records, avoid duplicates, track the progress of each student and perform timely follow-ups. This project replaces those methods with a centralised digital system.

The application contains three connected forms: Course, Student Enquiry and Follow-Up. An online enquiry form validates the data entered by the applicant, checks for duplicate email addresses and phone numbers, and sends an acknowledgement. Admission staff update the status of each enquiry through the stages New, Contacted, Interested, Applied, Admitted and Rejected, and record every follow-up with its date and outcome. Deluge scripts implement validation, notifications and a daily reminder schedule. Reports include list, Kanban and summary views, and an administrator dashboard displays status-wise and course-wise charts. Role-based access separates the tasks of the applicant, admission staff and administrator.

The report describes the problem, requirements, design, implementation, testing plan, deployment and security aspects of the system, together with an illustrative case study and the complete Deluge source code. The project demonstrates that a low-code platform can be used to build a practical institutional application quickly, with limited programming, and that it can be extended with SMS notifications, online applications and CRM integration.

Keywords: Zoho Creator; Deluge; Admission Enquiry; Low-Code; Workflow Automation; Dashboard

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Admission Enquiry Management System Using Zoho Creator
Application Domain	Education – college admissions
Student and Register No.	Harrish S – 2116241501501
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Department	B.Tech Artificial Intelligence and Machine Learning
Platform	Zoho Creator (cloud-based low-code platform)
Application Link	[Insert application link]
Roles	Applicant, Admission Staff, Administrator
Data Capture	3 forms: Course, Student Enquiry, Follow-Up
Reports	6 reports: lists, Kanban board and summaries
Dashboard	1 page with status-wise and course-wise charts
Automation	6 Deluge scripts including a daily reminder schedule
Integrations	Email notifications through Creator sendmail

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRM	Customer Relationship Management
DS	Deluge Script
ER	Entity Relationship
FK	Foreign Key
FR	Functional Requirement
ID	Identifier
PK	Primary Key
QR	Quick Response (code)
SMS	Short Message Service
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

1.1 Background

Every year, colleges receive hundreds of enquiries from prospective students through phone calls, walk-ins, social media and website forms. Each enquiry is a potential admission, and the way it is handled strongly influences the student's opinion of the institution. When enquiries are noted in paper registers or separate spreadsheets, details are easily lost, repeated or left without a reply.

Low-code platforms allow such operational problems to be solved quickly. Zoho Creator provides forms, a built-in database, reports, dashboards, workflows and the Deluge scripting language in one cloud environment

[1][2]. This project uses these features to build the Student Admission Enquiry Management System for an educational institution.

1.2 Problem Statement

Traditional admission enquiry management depends on paper forms and spreadsheets. This makes it difficult to maintain records, track the progress of each student and perform timely follow-ups. The main difficulties are listed below.

- Enquiry details are scattered across registers, sheets and messages.
- Duplicate or incomplete entries reduce the quality of data.
- Staff cannot easily see which students have been contacted and which are pending.
- Follow-up calls and emails are delayed or forgotten during the busy admission season.
- Preparing summary reports for management takes extra manual effort.

A structured and efficient digital solution is therefore required.

1.3 Objectives

- To collect and organise student enquiry details in a single centralised system.
- To track admission enquiries from the first contact to the final decision.
- To manage admission statuses such as New, Contacted, Interested, Applied, Admitted and Rejected.
- To automate routine tasks such as acknowledgement emails and follow-up reminders.
- To reduce paperwork and manual data entry.
- To provide reports and a dashboard that support quick decisions.

1.4 Scope of the Project

The project covers the enquiry stage of the admission process. It does not replace the complete college admission or student information system.

Table 1.1: Scope of the Project

In Scope	Out of Scope
Online enquiry form for prospective students	Online payment of application fees
Secure login for admission staff and administrators	Student record management after admission
Enquiry tracking, status updates and follow-up records	Document verification and entrance examination modules
Automated email notifications	Integration with the university or government portals
Reports and a summary dashboard	Native SMS or WhatsApp messaging

1.5 Justification for a Low-Code Approach

A custom web application would need a separate database, server, user interface, security layer and hosting arrangement. Zoho Creator supplies these elements as ready services, so the developer concentrates on the admission process itself. Forms and reports are created visually, and Deluge scripts are added only where

business rules need them. This shortens development time, reduces cost and allows the college staff to modify the application later without deep programming knowledge [1].

1.6 Organisation of the Report

Chapters 2 to 4 present the system study, requirement analysis and design. Chapters 5 to 7 describe the implementation, testing, deployment, security and governance. Chapter 8 discusses the results and Chapter 9 gives the conclusion and future enhancements. The report ends with references and appendices that contain a case study, the Deluge source code and a user manual.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In many colleges the admission office records each enquiry manually. A staff member writes the name, phone number and course of interest in a register or enters it into a spreadsheet. Phone calls and social media messages are noted separately. Follow-up depends on the memory and diligence of individual staff members.

2.2 Limitations of the Existing System

Table 2.1: Limitations of the Existing System

Area	Limitation
Data storage	Records are spread over registers, spreadsheets and messages; there is no single source of truth.
Data quality	No validation; duplicate, incomplete and incorrectly formatted entries are common.
Tracking	The status of an enquiry cannot be seen quickly; progress is not recorded in a standard way.
Follow-up	No reminders; calls depend on individual memory and may be missed.
Communication	Students do not receive an acknowledgement after submitting an enquiry.
Reporting	Counts by course, status or source must be prepared manually.
Security	Paper registers and shared files have no role-based access or audit trail.

2.3 Low-Code versus Traditional Development

Table 2.2: Comparison of Low-Code and Traditional Development for the Proposed Application

Aspect	Traditional Development	Zoho Creator (Low-Code)
Development effort	Separate design of database, back end and front end	Forms, reports and workflows built visually
Infrastructure	Server, hosting and maintenance required	Cloud hosted by the platform
Change management	Code changes and redeployment	Edit forms and scripts in the builder
Security	Must be coded and tested	Role-based access and encryption provided
Mobile access	Separate mobile application	Responsive web and mobile apps generated
Skills needed	Experienced programmers	Basic logic skills; Deluge for rules

2.4 Literature Survey

The following sources were studied to understand the platform, the scripting language and the software engineering principles applied in the project.

Table 2.3: Summary of Sources Studied

Ref.	Source	Relevance to the Project
[1]	Zoho Creator official documentation	Forms, reports, pages, workflows, roles and deployment features of the platform.
[2]	Deluge scripting guide	Syntax for fetch, conditions, loops, validation and sendmail tasks used in the rules.
[3]	I. Sommerville, Software Engineering	Requirements engineering, system modelling and testing principles.
[4]	R. S. Pressman, Software Engineering: A Practitioner's Approach	Process models, requirement analysis and test case design.
[5]	R. Elmasri and S. B. Navathe, Fundamentals of Database Systems	Entity relationships, keys and normalisation used in the data model.

2.5 Emerging Trends in Low-Code Development

Low-code platforms are increasingly used by educational and business organisations to digitise processes quickly. Current trends include browser-based builders, reusable components, integration with other cloud

applications, built-in analytics and the use of artificial intelligence to assist developers. These trends make it practical for a student project to produce a working application that closely resembles a real institutional tool.

2.6 Proposed System

The proposed system is a Zoho Creator application with an online enquiry form, a centralised database, status tracking, a follow-up register, automated emails and reports. Students submit enquiries once; the system validates the data, checks for duplicates, sends an acknowledgement and notifies the admission staff. Staff update the status and record follow-ups, while the administrator monitors the dashboard.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Main Responsibilities
Applicant	Prospective student or parent	Submits the enquiry and receives an acknowledgement.
Admission Staff	Admission office counsellor	Views enquiries, updates status, records follow-ups.
Administrator	Admission head or system owner	Manages users, courses, reports and access rights.

3.2 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement
FR-01	The system shall provide an online enquiry form with mandatory field validation.
FR-02	The system shall store every enquiry in a central database with an automatic Enquiry ID.
FR-03	The system shall warn the user when the same email or phone number already exists.
FR-04	The system shall send an acknowledgement email to the student after submission.
FR-05	The system shall notify admission staff when a new enquiry is received.
FR-06	The system shall allow staff to update the admission status of an enquiry.
FR-07	The system shall record follow-up date, mode, remarks and next follow-up date.
FR-08	The system shall send a daily reminder for follow-ups due on that day.
FR-09	The system shall provide search and filter facilities on enquiry reports.
FR-10	The system shall display a dashboard with status-wise and course-wise summaries.
FR-11	The system shall restrict data access according to user roles.

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

Attribute	Requirement
Usability	Simple forms and menus that staff can use with minimal training.
Performance	Forms and reports should load within a few seconds under normal use.
Security	Login is required for staff; role-based permissions protect personal data.
Availability	Cloud hosting provides access from any location with internet.
Maintainability	Fields, status values and workflows can be changed in the builder.
Portability	The application works on common browsers and on mobile devices.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Requirement
Platform	Zoho Creator (cloud based)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Client device	Computer, tablet or smartphone with internet access
Browser	Current version of Chrome, Edge or Firefox
Users	Applicant, Admission Staff, Administrator

3.5 Use Case Model

Figure 3.1 shows the actors and the main use cases of the system. The applicant interacts only with the enquiry form and the acknowledgement; staff handle the enquiry life cycle; the administrator controls configuration and reports.

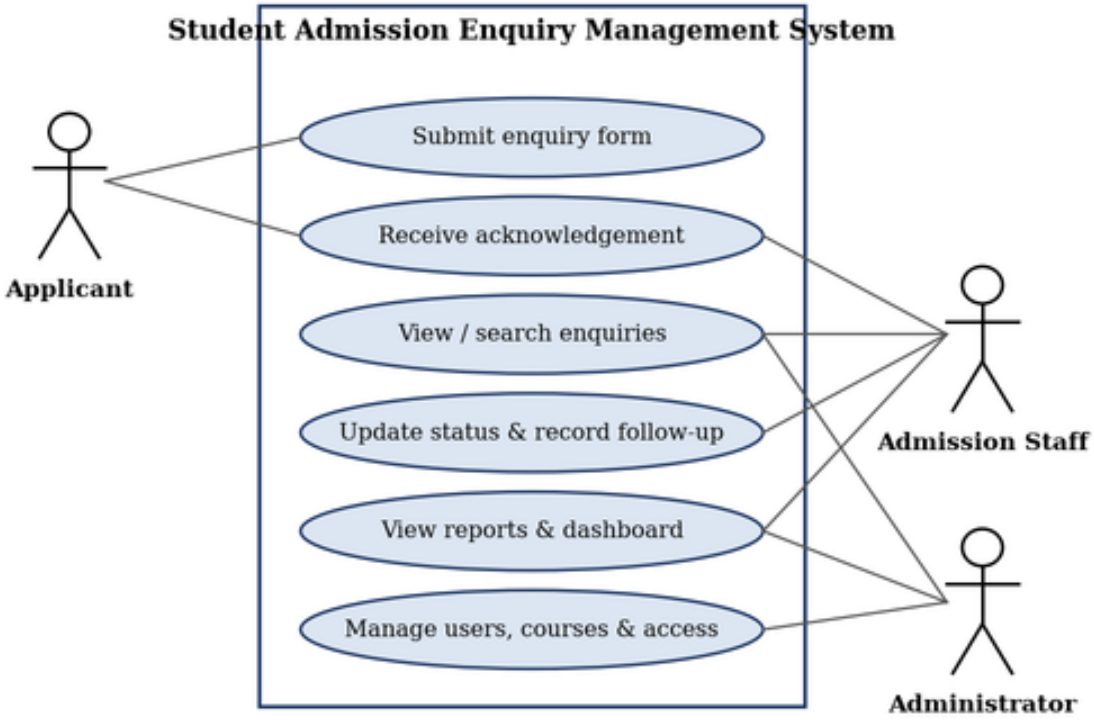


Figure 3.1: Use Case Diagram of the Admission Enquiry System

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The application follows the standard Zoho Creator architecture. Users access the forms and reports through a browser or the mobile app. Workflows and Deluge scripts execute on the platform whenever a record is added, edited or scheduled. All records are stored in the Creator database, and notification emails are sent through the built-in mail service.

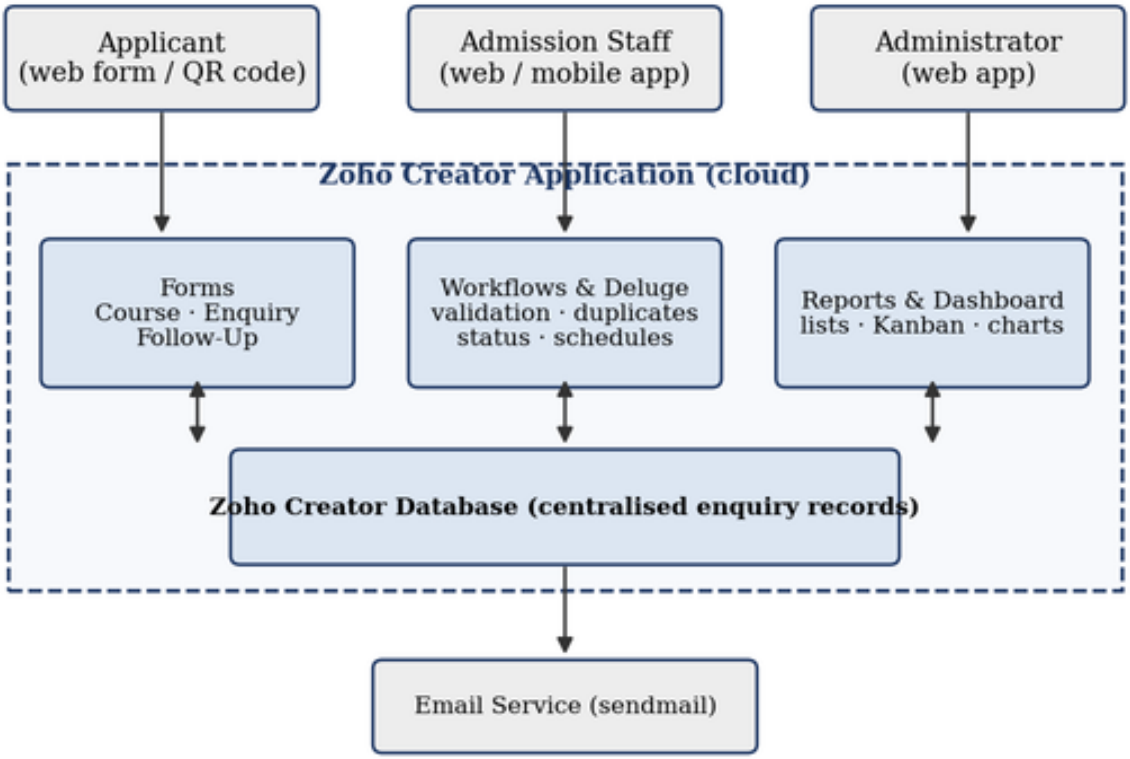


Figure 4.1: System Architecture

4.2 Data Model

The data model has three forms. A course can have many enquiries, and an enquiry can have many follow-up records. These one-to-many relationships are implemented as lookup fields.

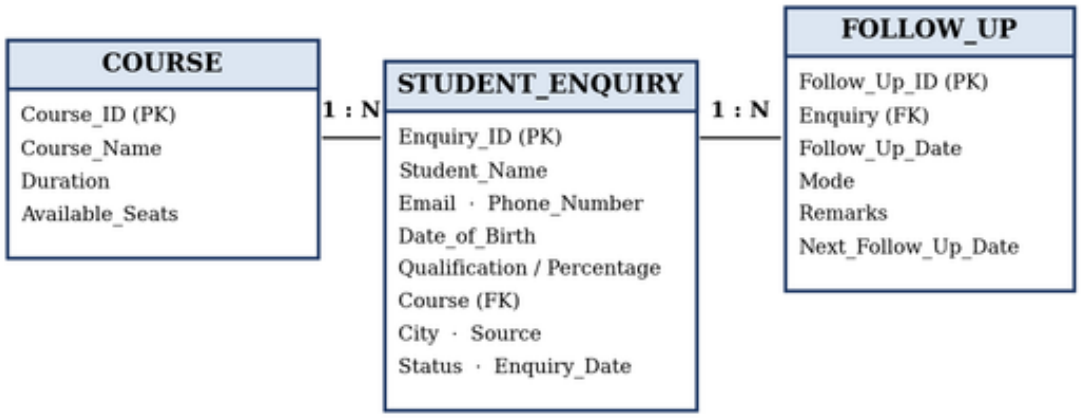


Figure 4.2: Entity Relationship Diagram

Table 4.1: Entities and Relationships

Entity	Primary Key	Related To	Relationship
Course	Course_ID	Student Enquiry	One course – many enquiries
Student Enquiry	Enquiry_ID	Follow-Up	One enquiry – many follow-ups
Follow-Up	Follow_Up_ID	Student Enquiry	Many follow-ups – one enquiry

4.3 Form Design

Table 4.2: Student Enquiry Form Specification

Field	Type	Validation / Note
Enquiry ID	Auto Number	Generated by the system
Student Name	Name	Mandatory
Email	Email	Mandatory; valid format; duplicate check
Phone Number	Phone	Mandatory; digits only; duplicate check
Date of Birth	Date	Must be a past date
Last Qualification	Text	Mandatory
Percentage	Decimal	Between 0 and 100
Course Interested	Lookup (Course)	Mandatory
City	Text	Optional
Source of Enquiry	Dropdown	Website, Social Media, Friend, Advertisement
Message	Multi-line Text	Optional
Status	Dropdown	Default “New”; set by staff afterwards
Enquiry Date	Date-Time	Added automatically

Table 4.3: Course and Follow-Up Form Specification

Form	Fields
Course	Course Name, Duration, Available Seats
Follow-Up	Enquiry (lookup), Follow-Up Date, Mode (Call / Email / Visit), Remarks, Next Follow-Up Date

4.4 Process and Workflow Design

Each enquiry moves through a defined set of statuses, as shown in Figure 4.3. A student who loses interest or is not eligible can be moved to the Not Interested or Rejected status. Every status change is stored so that management can analyse progress.

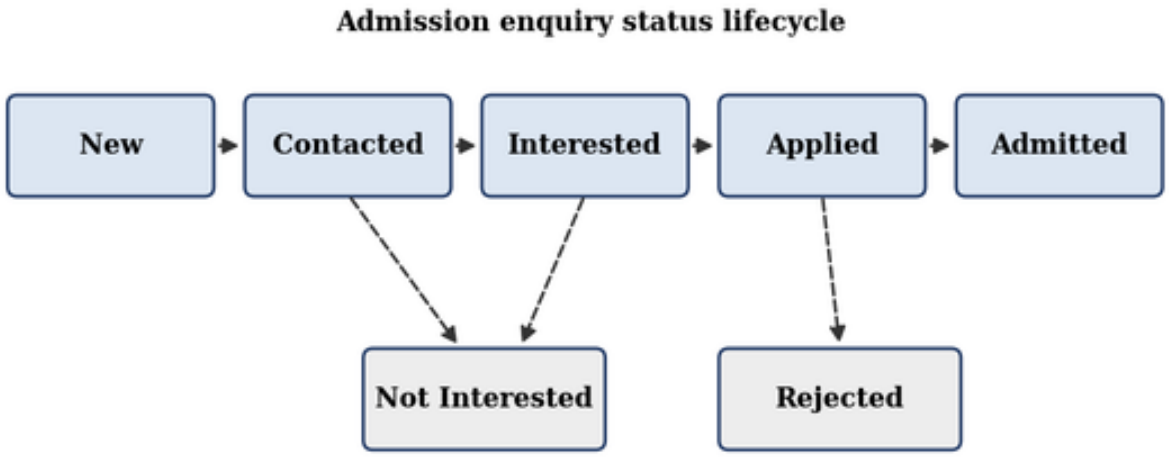


Figure 4.3: Enquiry Status Life Cycle

4.5 User Interface and Navigation Design

The application menu is arranged by task. The public enquiry form is available without login. After login, staff see the Enquiries list, Follow-Ups, Pending Follow-Ups and the Dashboard. The administrator additionally sees the Courses form and user settings.

Table 4.4: Navigation Structure

Menu Item	Component	Visible To
Enquiry Form	Form	Applicant, Staff, Administrator
All Enquiries	List report with search and filter	Staff, Administrator
Enquiry Board	Kanban report grouped by status	Staff, Administrator
Follow-Ups	Form and report	Staff, Administrator
Pending Follow-Ups	Filtered report	Staff, Administrator
Dashboard	Page with charts and counts	Administrator
Courses	Form and report	Administrator

4.6 Access Control Design

Table 4.5: Role Permissions

Component	Applicant	Staff	Administrator
Enquiry Form	Add	Add / Edit	Full
Enquiry Reports	—	View / Edit	Full
Follow-Up Form	—	Add / Edit	Full
Courses	—	View	Full
Dashboard	—	View	Full

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

A new application named “Student Admission Enquiry Management System” was created in Zoho Creator. The date format, time zone and application link name were set from the application settings. The three forms, their reports, the dashboard page and the workflows were then added in sequence.

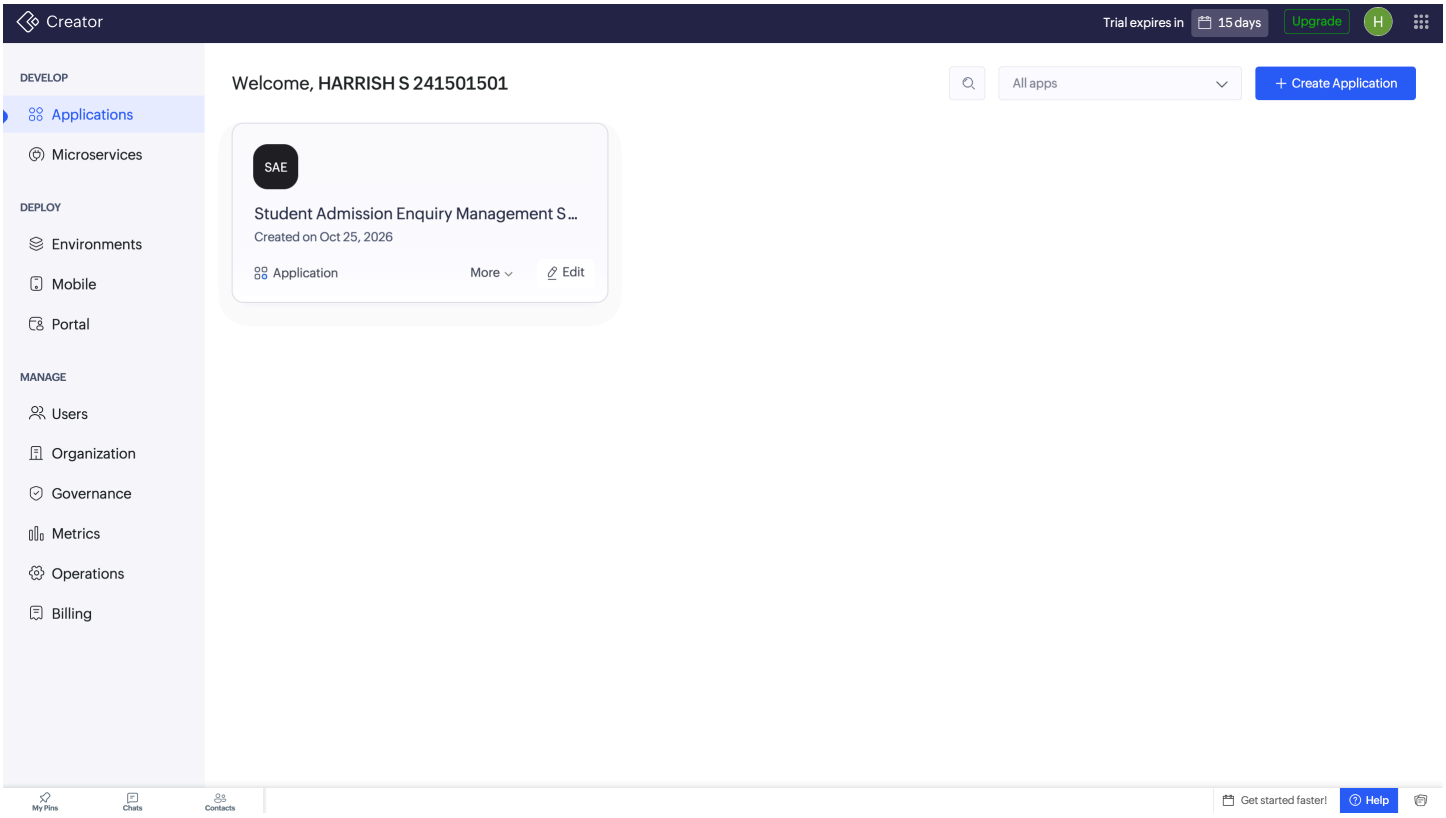


Figure 5.1: Application Navigation Menu

5.2 Forms and Data Capture

The Course form stores the programmes offered and the seats available. The Student Enquiry form captures the student's details and is linked to the Course form through a lookup field. The Follow-Up form is linked to the enquiry and stores each contact attempt. Default values and field rules are applied when a form loads, for example the status is set to "New" and the enquiry date is filled automatically.

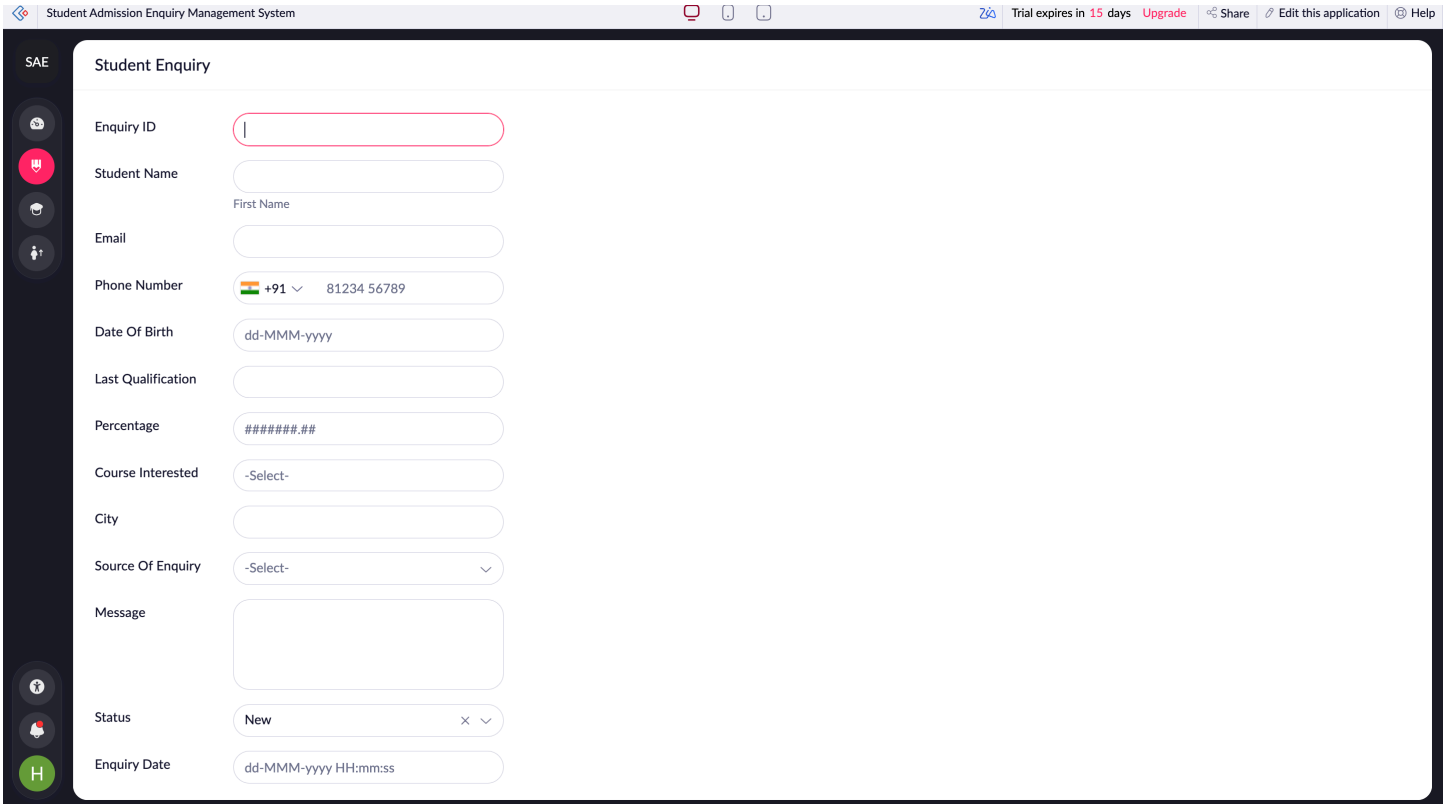


Figure 5.2: Student Enquiry Form

5.3 Reports

Table 5.1: Reports Implemented

Report	Type	Source Form	Purpose
All Enquiries	List	Student Enquiry	Search, filter and edit all enquiries
Enquiry Board	Kanban	Student Enquiry	Visual view of enquiries by status
Course-wise Enquiries	Summary	Student Enquiry	Count of enquiries for each course
Enquiries by Status	Summary	Student Enquiry	Count of enquiries in each status
Pending Follow-Ups	List	Follow-Up	Follow-ups due today or overdue
Courses	List	Course	Programme and seat details

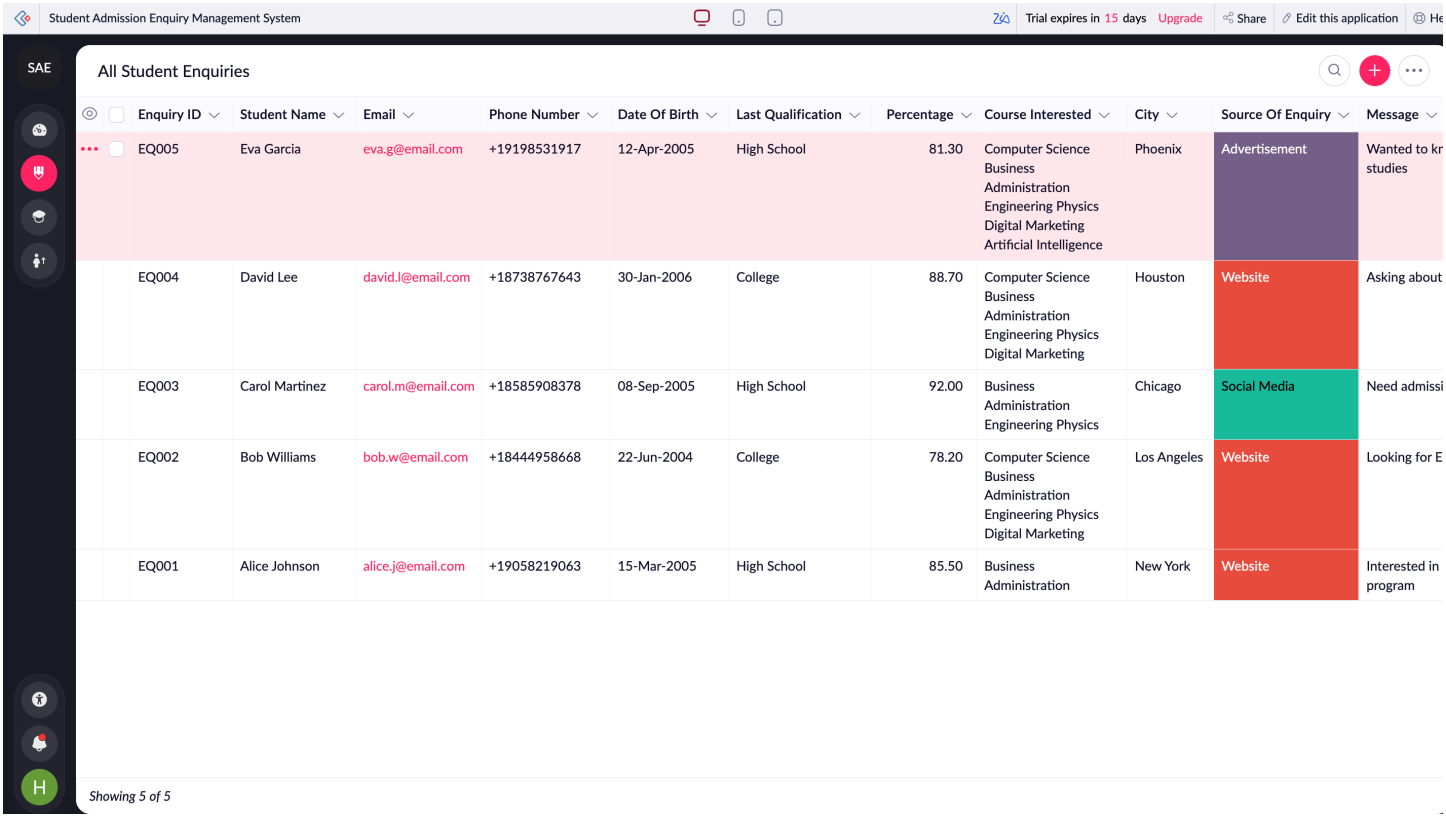


Figure 5.4: All Enquiries List Report

5.4 Pages and Dashboards

The Admission Dashboard is a Creator page that shows the total number of enquiries, a status-wise chart, a course-wise chart and a monthly enquiry trend. It also contains links to the important forms and reports so that the administrator can reach them in one click.

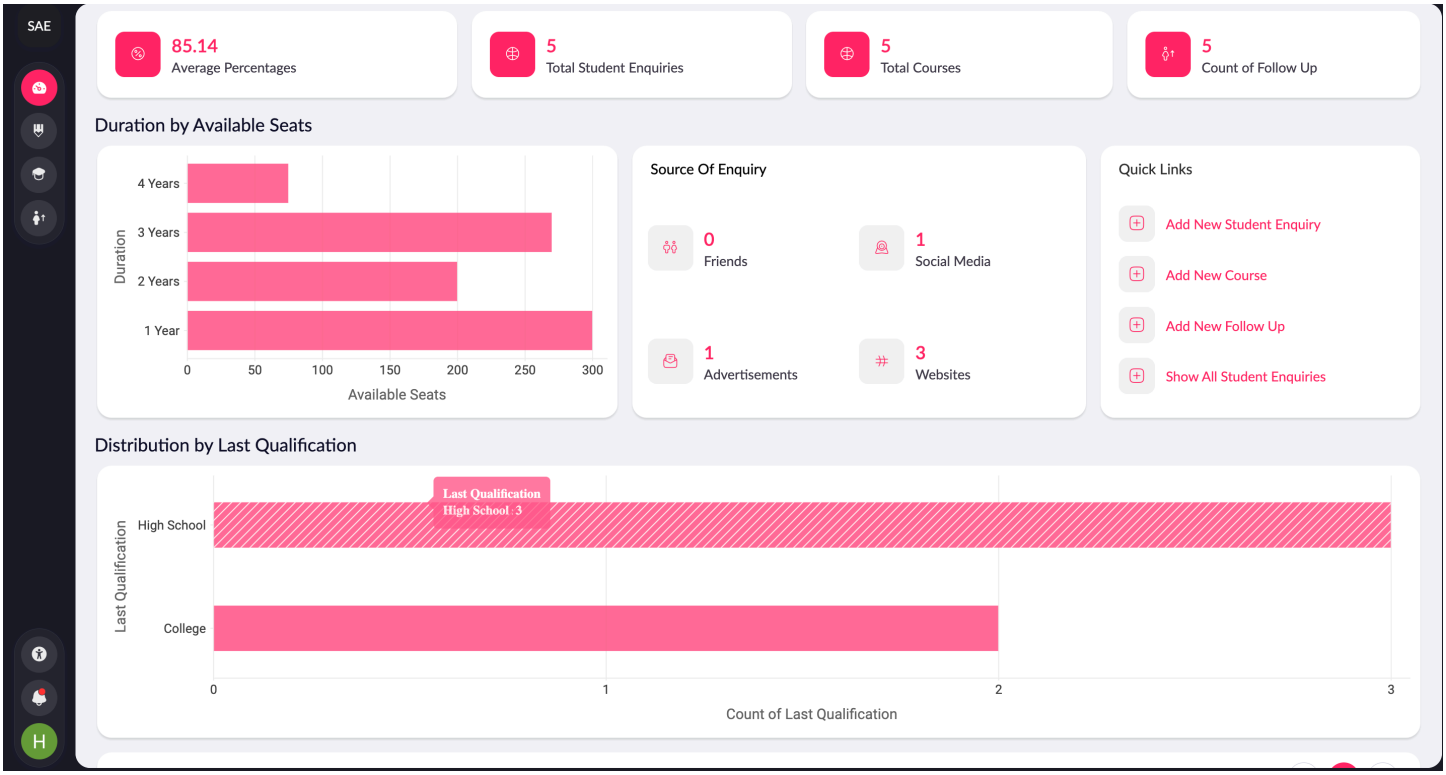


Figure 5.5: Admission Dashboard

5.5 Deluge Scripting

Deluge scripts implement the business rules that cannot be configured through settings. Table 5.2 summarises the scripts, and Codes 5.1 and 5.2 show two important examples. The complete source code is given in Appendix B.

Table 5.2: Deluge Scripts Implemented

Script	Location and Trigger	Purpose
DS-01	Student Enquiry – On Validate	Check duplicate email or phone number
DS-02	Student Enquiry – On Success	Send acknowledgement email and staff alert
DS-03	Student Enquiry – On Load	Set default status and enquiry date
DS-04	Student Enquiry – On Edit (status field)	Send status update email to the student
DS-05	Follow-Up – On Validate	Ensure next follow-up date is after follow-up date
DS-06	Schedule – Daily 08:00	Send reminders for follow-ups due today

Code 5.1: Duplicate Enquiry Check (On Validate)

```
dup = Student_Enquiry[(Email == input.Email
        || Phone_Number == input.Phone_Number)
        && ID != input.ID];
if(dup.count() > 0)
{
    alert "This email or phone number is already registered.";
    cancel submit;
}
```

Code 5.2: Acknowledgement Email (On Success)

```
sendmail
[
    from :zoho.adminuserid
    to :input.Email
    subject :"Admission Enquiry Received - " + input.Enquiry_ID
    message :"Dear " + input.Student_Name + ", thank you for "
        + "your enquiry. Our admission team will contact you."
]
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow	Type	Trigger	Action	Script
Enquiry validation	Form	Validate	Duplicate and format checks	DS-01
Acknowledgement	Form	Success	Email to student and staff alert	DS-02
Default values	Form	Load	Status = New; date filled	DS-03
Status notification	Form	Edit	Email when status changes	DS-04
Follow-up check	Form	Validate	Date comparison	DS-05
Daily reminder	Schedule	Daily 08:00	Reminder mail to staff	DS-06

5.7 Integrations

Table 5.4: Integrations Used

Integration	Service	Purpose
Email notifications	Creator sendmail task	Acknowledgements, alerts and reminders
QR code / web link	Creator public form link	Allows applicants to open the form from posters and social media
Payment, SMS and maps	Not used	Outside the scope of this project

5.8 User Management and Role-Based Access

Three roles are defined: Administrator, Admission Staff and Applicant. The Administrator has full access. Admission Staff can add and edit enquiries and follow-ups but cannot change courses or users. The enquiry form is published as a public form so that applicants can submit it without an account. Permissions are applied to forms, reports and pages, so that data is hidden by the platform and not merely by hiding fields.

5.9 AI and Analytics Features

The dashboard charts provide status-wise, course-wise and month-wise analysis. AI assistance was used during development to suggest Deluge code and to review the report. All generated code was read, understood and tested by the student before use. No prediction or sentiment-analysis feature is claimed.

5.10 Mobile and Portal Access

Zoho Creator automatically makes the application available in a mobile-friendly layout and through the Creator mobile app. Staff can therefore record follow-up outcomes immediately after a call.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at three levels. Unit testing checks each Deluge script and validation rule. Integration testing checks that forms, reports and workflows work together. User acceptance testing checks that the system supports the real tasks of admission staff. Test data consisting of sample students, courses and follow-ups is entered through the application forms.

6.2 Test Cases

Table 6.1 lists the test cases and the expected results. The actual result and status are recorded when each case is executed in the application.

Table 6.1: Test Cases

ID	FR	Test	Input / Steps	Expected Result	Status
TC-01	FR-01	Valid enquiry	Fill all fields correctly and submit	Record saved with Enquiry ID	
TC-02	FR-01	Mandatory field	Leave name or email blank	Error message; record not saved	
TC-03	FR-01	Invalid email	Enter “abc@” as email	Validation message shown	
TC-04	FR-03	Duplicate email	Submit an email that already exists	Duplicate warning; submit cancelled	
TC-05	FR-04	Acknowledgement	Submit a valid enquiry	Email sent to the student	
TC-06	FR-05	Staff alert	Submit a valid enquiry	Alert received by staff	
TC-07	FR-06	Status change	Change status to Admitted	Status saved; student notified	
TC-08	FR-07	Follow-up dates	Next date earlier than follow-up date	Error message shown	
TC-09	FR-08	Daily reminder	Follow-up due today; run schedule	Reminder email sent	
TC-10	FR-10	Dashboard	Open dashboard after adding records	Counts match the reports	
TC-11	FR-11	Role access	Login as staff; open Courses	Only view access allowed	

6.3 Debugging Deluge Scripts

Deluge errors are found with the built-in script editor, which highlights syntax errors, and with the info statement, which prints variable values while a workflow runs. Common problems and the usual remedies are given in Table 6.2.

Table 6.2: Common Deluge Problems and Remedies

Problem	Likely Cause	Remedy
Script does not run	Workflow trigger or condition incorrect	Check the event (validate, success, edit) and the condition
Fetched value is empty	Criteria compares wrong field or case	Verify field link names and print the result with info
Email not received	Wrong address or mail blocked	Test with a known address and check spam folder
Duplicate check blocks edits	Record compares with itself	Exclude the current record using ID != input.ID
Schedule does not fire	Start date or time zone not set	Review the schedule configuration

6.4 Defect Log

Defects found while executing the test cases are recorded in the format shown in Table 6.3 together with their correction.

Table 6.3: Defect Log

No.	Test ID	Defect Description	Correction	Status
1				
2				
3				

6.5 User Acceptance Testing

Admission staff are asked to perform their normal tasks: register a walk-in enquiry, change the status, record a follow-up and view the dashboard. The application is accepted when these tasks are completed without assistance and the staff confirm that the screens are clear.

Table 6.4: User Acceptance Checklist

Task	Performed By	Accepted
Register a walk-in enquiry	Admission staff	
Change the status of an enquiry	Admission staff	
Record a follow-up and next date	Admission staff	
View reports and dashboard	Administrator	

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides a development workspace and a production environment. The application is built and tested in the development workspace with sample data. After the test cases are completed, the application is published, the public form link is shared, and staff accounts are created. Changes after publishing are first made in the development workspace and then moved to production.

Table 7.1: Deployment Checklist

Step	Activity
1	Complete testing with sample data
2	Remove test records from the database
3	Create staff users and assign roles
4	Publish the enquiry form and generate the public link and QR code
5	Confirm that email notifications are enabled
6	Train the admission staff

7.2 Performance Optimisation

- Reports show only the required columns and use filters, for example Pending Follow-Ups.
- Scripts fetch records with specific criteria instead of loading entire forms.
- Dashboard charts use summary reports rather than detailed lists.
- The schedule runs once a day, outside working hours, to avoid load.

7.3 Security Features

Table 7.2: Security Features

Feature	Description
Authentication	Staff and administrators must sign in with an account.
Role-based access	Permissions on forms, reports and pages follow user roles.
Data validation	Mandatory fields and format rules prevent invalid data.
Data protection	The platform encrypts data in transit and takes regular backups.
Public form control	The public form can add records but cannot view existing data.

7.4 Governance and Administration

The administrator is responsible for creating and removing users, reviewing permissions and maintaining the list of courses. Applicant data must be used only for admission purposes, and access should be removed when a staff member leaves. Records that are no longer needed should be archived or deleted according to the institution’s data policy.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The following screens show the working application. They are captured from the live application after entering sample data.

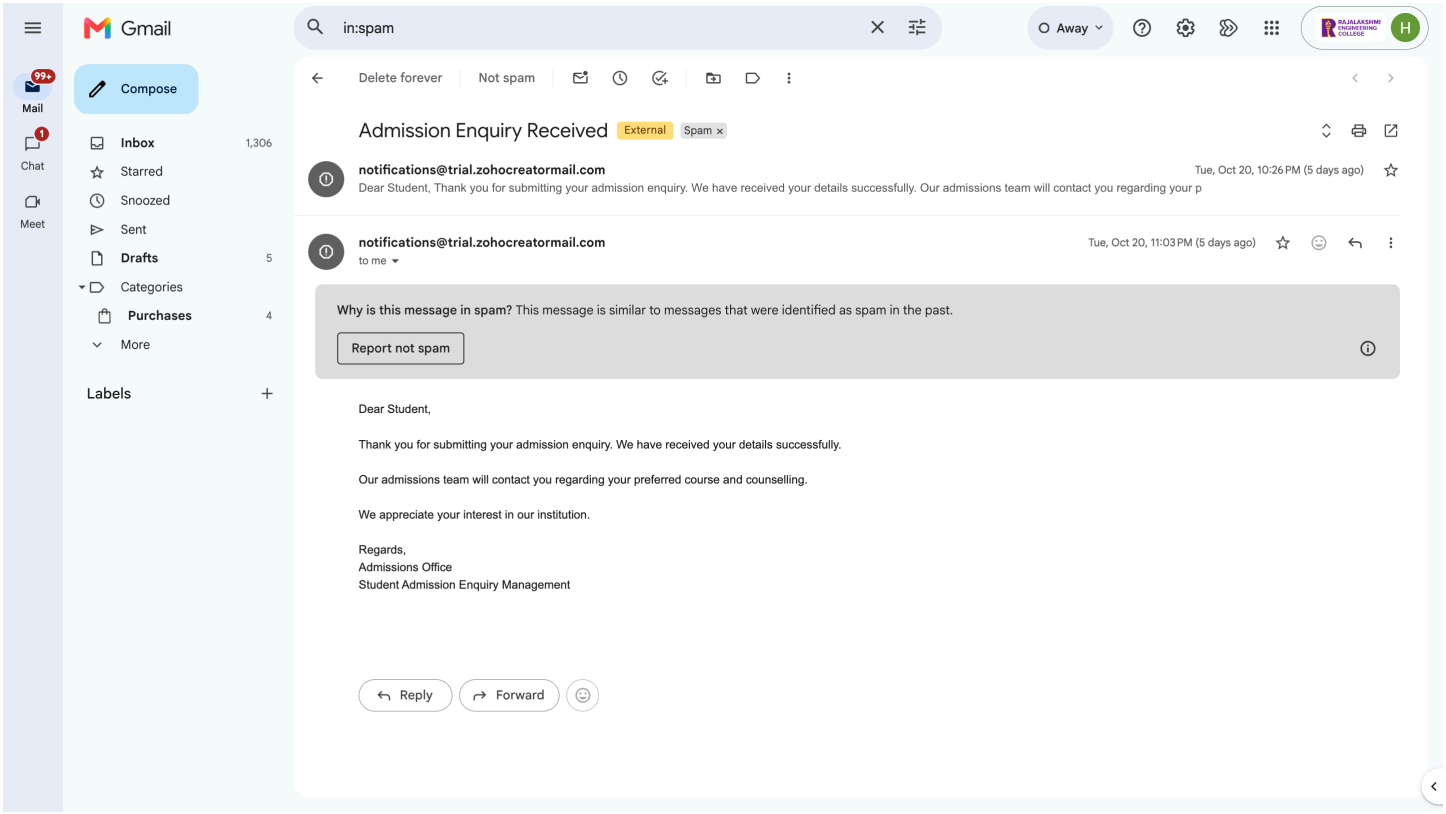


Figure 8.1: Acknowledgement Email

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Achieved Through
Collect and organise enquiry details	Student Enquiry form with validation and a central database
Track enquiries to the final decision	Status field, Enquiry Board and Follow-Up form
Manage admission statuses	Dropdown statuses with notification on change
Automate routine tasks	Acknowledgement emails, staff alerts and the daily reminder schedule
Reduce paperwork and manual entry	Online form and public link replace paper registers
Provide reports and dashboard	Six reports and the Admission Dashboard

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed System

Feature	Existing System	Proposed System
Record keeping	Registers and spreadsheets	Single centralised database
Data validation	None	Mandatory, format and duplicate checks
Acknowledgement	None	Automatic email
Follow-up	Depends on memory	Recorded with reminders
Status tracking	Not available	Status field and Kanban board
Reports	Manual preparation	Instant reports and dashboard
Access control	None	Role-based permissions

8.4 Limitations of the Low-Code Approach

- The application depends on the platform, its subscription plan and its usage limits.
- Very specialised features may need Deluge or integration with other services.
- An internet connection is necessary for all users.
- Vendor-specific skills are needed to maintain the application.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of this application was created without writing code, by arranging forms, reports and settings. Deluge scripting was needed for duplicate checks, notifications and the daily schedule, which places the project in the low-code range. A pro-code extension, such as an external service or custom widget, could be added in future if the institution needs functions that the platform does not provide.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Admission Enquiry Management System replaces manual enquiry handling with a simple digital solution built on Zoho Creator. It stores data in one place, validates entries, tracks each enquiry through defined statuses and automates routine communication. The project shows that a low-code platform can be used to build a useful institutional application quickly and with limited programming. With the planned enhancements the system can grow into a complete admission management solution.

9.2 Future Enhancements

- SMS and WhatsApp notifications for enquiries and reminders.
- Online application form with document upload.
- Online payment of application fees.
- Integration with Zoho CRM for lead management.

- Chatbot to answer common admission questions.
- Advanced analytics to study enquiry sources and conversion rates.

9.3 Learning Outcomes and Career Relevance

Through this project the student learned to analyse a real process, design a relational data model, build forms and reports, write Deluge scripts, configure workflows and roles, and test and document an application. These skills are relevant to roles such as low-code developer, business application analyst and software tester.

REFERENCES

[1] Zoho Corporation, "Zoho Creator Help Documentation," https://www.zoho.com/creator/help/.

[2] Zoho Corporation, "Deluge Scripting Language Guide," https://www.zoho.com/deluge/help/.

[3] I. Sommerville, Software Engineering, 10th ed. Pearson Education, 2016.

[4] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 8th ed. McGraw-Hill Education, 2014.

[5] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed. Pearson Education, 2016.

APPENDIX A – CASE STUDY: ADMISSION ENQUIRY HANDLING AT A SAMPLE COLLEGE

Note: This is an illustrative case study. The college name and figures are sample data used to demonstrate how the system works.

A.1 Background

ABC Arts and Science College offers ten undergraduate programmes. Every year the admission office receives about 3,000 enquiries between April and July through walk-ins, phone calls, the college website and social media.

A.2 Challenges Faced

- Enquiries from different sources were recorded in different places.
- About 15% of entries were duplicates or had wrong phone numbers.
- Staff could not easily see which students had been contacted.
- Follow-up calls were missed during the peak admission season.
- Preparing a weekly report for the principal took nearly one full day.
- Students received no confirmation after submitting an enquiry.

A.3 Solution Implemented

The college adopted the Student Admission Enquiry Management System. The enquiry form was placed on the website and shared through a QR code on posters and social media. Walk-in and phone enquiries were entered by staff using the same form. Each enquiry received a status, automated emails were sent, and a dashboard was provided to the principal and the admission head.

A.4 Implementation Steps

Table A.1: Implementation Steps of the Case Study

Step	Activity
1	Studied the existing admission process and listed required data fields
2	Created the Student Enquiry, Course and Follow-Up forms
3	Added validation rules and the duplicate check
4	Built workflows for email notifications and follow-up reminders
5	Created reports and the dashboard
6	Assigned roles and permissions to staff
7	Tested with sample data and trained the admission staff
8	Went live at the start of the admission season

A.5 Sample Enquiry Journey

1. A student, Priya, scans the QR code and submits an enquiry for the B.Sc. Computer Science course.
2. She instantly receives an acknowledgement email with her Enquiry ID.
3. The system alerts the staff member responsible.
4. The staff member calls Priya the next day, records the call in the Follow-Up form and changes the status to Interested.
5. A reminder is sent three days later for the next follow-up.
6. Priya applies and the status becomes Applied and then Admitted.
7. The dashboard updates the course-wise and status-wise counts automatically.

A.6 Results (Sample Comparison)

Table A.2: Sample Results of the Case Study

Measure	Before the System	After the System
Time to record one enquiry	5–6 minutes	1–2 minutes
Duplicate or wrong entries	About 15%	Below 2%
Follow-ups completed on time	About 50%	About 90%
Time to prepare weekly report	1 day	A few seconds (live dashboard)
Student acknowledgement	None	Immediate email

A.7 Lessons Learned

- Clear data fields and validation rules at the start prevent data quality problems later.
- Staff training is as important as the application itself.
- Starting with a simple version and adding features later works well on a low-code platform.

APPENDIX B – COMPLETE DELUGE SOURCE CODE

Code B.1: DS-01 Duplicate Check (Student Enquiry – On Validate)

```
dup = Student_Enquiry[(Email == input.Email
        || Phone_Number == input.Phone_Number)
        && ID != input.ID];
if(dup.count() > 0)
{
    alert "This email or phone number is already registered.";
    cancel submit;
}
```

Code B.2: DS-02 Acknowledgement and Staff Alert (On Success)

```
sendmail
[
    from :zoho.adminuserid
    to :input.Email
    subject :"Admission Enquiry Received - " + input.Enquiry_ID
    message :"Dear " + input.Student_Name + ", thank you for "
        + "your enquiry. Our admission team will contact you."
]
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"New enquiry: " + input.Student_Name
    message :"Course: " + input.Course_Interested.Course_Name
]
```

Code B.3: DS-03 Default Values (On Load)

```
input.Status = "New";
input.Enquiry_Date = zoho.currenttime;
hide Enquiry_Date;
```

Code B.4: DS-05 Follow-Up Date Check (On Validate)

```
if(input.Next_Follow_Up_Date != null
   && input.Next_Follow_Up_Date <= input.Follow_Up_Date)
{
    alert "Next follow-up date must be after follow-up date.";
    cancel submit;
}
```

Code B.5: DS-06 Daily Follow-Up Reminder (Schedule)

```
today = zoho.currentdate;
due = Follow_Up[Next_Follow_Up_Date == today];
for each f in due
{
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"Follow-up due: " + f.Enquiry.Student_Name
        message :"Please contact " + f.Enquiry.Student_Name
            + " (" + f.Enquiry.Phone_Number + ")."
    ]
}
```

APPENDIX C – USER MANUAL

C.1 Applicant

1. Open the enquiry link or scan the QR code.
2. Enter the name, email, phone number, qualification and course of interest.
3. Click Submit. A confirmation email with the Enquiry ID is received.

C.2 Admission Staff

1. Sign in to Zoho Creator and open the application.
2. Open All Enquiries to search or filter by name, course or status.
3. Open an enquiry, change its status and save.
4. Open the Follow-Up form, select the enquiry and record the call, remarks and next date.
5. Check Pending Follow-Ups every morning.

C.3 Administrator

1. Add or edit courses and available seats in the Courses form.
2. Create staff users and assign roles from user management.
3. Open the Admission Dashboard to review totals and charts.